Equation of motion

$$x[k+1] = x[k] + a_x[k]\{f_{dx}[k] + f_T[k]\},$$

Motion error

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{f_{dx}[k] + f_T[k]\},$$

$\delta x[k] = x_d[k] - x[k]$, $x_d[k]$ is the desired motion trajectory, and $\Delta x_d[k] = x_d[k+1] - x_d[k]$ is the desired motion step size.

Measurement

(d-step delay)

$$\delta x_m[k] = \delta x[k-d] + n_x[k],$$

Control objective: $\delta x[k+1] = \lambda_c \cdot \delta x[k]; \quad 0 \le \lambda_c < 1$

Ideal control effort (no measurement delay):

$$\bar{f}_{dx}[k] = a_x^{-1}[k]\{\Delta x_d[k] + (1-\lambda_c)\,\boldsymbol{\delta x[k]}\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{f}_{dx}[k] + f_T[k]\} = \lambda_c \cdot \delta x[k] - a_x[k] f_T[k]$$

d-step measurement delay:

$$\boldsymbol{\delta x[k]} = \delta x[k-d] + \sum_{i=1}^{d} \Delta x_d[k-i] - \sum_{i=1}^{d} a_x[k-i]\{f_{dx}[k-i] + f_T[k-i]\}$$

Ideal control effort (with d-step measurement delay):

$$\bar{\bar{f}}_{dx}[k] = a_x^{-1}[k]\Big\{ \underbrace{\Big[\Delta x_d[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} \Delta x_d[k-i]\Big]}_{\Delta x_d^d[k]} + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]\Big\}$$

Motion error:

$$\delta x[k+1] = \delta x[k] + \Delta x_d[k] - a_x[k]\{\bar{\bar{f}}_{dx}[k] + f_T[k]\} = \lambda_c \cdot \delta x[k] - \underbrace{\Big\{a_x[k] f_T[k] + (1-\lambda_c)\textstyle\sum_{i=1}^{d} a_x[k-i] f_T[k-i]\Big\}}_{\delta x_T^d[k]}$$

Implementable control effort

$$\boxed{f_{dx}[k] = \hat{a}_x^{-1}[k]\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x_m[k] - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}}$$

Motion error

$$\delta x[k+1] = \lambda_c \cdot \delta x[k] - a_x[k]\big[f_{dx}[k] - \bar{\bar{f}}_{dx}[k]\big] - \delta x_T^d[k]$$

$$a_x[k]\, \bar{\bar{f}}_{dx}[k] = \Delta x_d^d[k] + (1-\lambda_c)\Big[\delta x[k-d] - \textstyle\sum_{i=1}^{d} a_x[k-i] f_{dx}[k-i]\Big]$$

$$a_x[k]\, f_{dx}[k] = \big(1 + e_{a_x}[k]\, \hat{a}_x^{-1}[k]\big)\Big\{\Delta x_d^d[k] + (1-\lambda_c)\Big[\big(\delta x[k-d] + n_x[k]\big) - \textstyle\sum_{i=1}^{d} \hat{a}_x[k-i] f_{dx}[k-i]\Big]\Big\}$$

$$a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] = e_{a_x}[k]\, f_{dx}[k] + (1-\lambda_c) n_x[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} \big(a_x[k-i] - \hat{a}_x[k-i]\big) f_{dx}[k-i]$$

Parameter model:

$$\begin{cases} a_x[k+1] = a_x[k] + \delta a_x[k] \\ \delta a_x[k+1] = \delta a_x[k] \end{cases}$$

Parameter estimator:

$$\begin{cases} \hat{a}_x[k+1] = \hat{a}_x[k] + \delta \hat{a}_x[k] \\ \delta \hat{a}_x[k+1] = \delta \hat{a}_x[k] \end{cases}$$

Parameter estimation error:

$$\begin{cases} e_{a_x}[k] = a_x[k] - \hat{a}_x[k] \\ e_{\delta a_x}[k] = \delta a_x[k] - \delta \hat{a}_x[k] \end{cases}$$

Estimation error dynamics:

$$\begin{cases} e_{a_x}[k+1] = e_{a_x}[k] + e_{\delta a_x}[k] \\ e_{\delta a_x}[k+1] = e_{\delta a_x}[k] \end{cases}$$

$$a_x[k-i] - \hat{a}_x[k-i] \approx e_{a_x}[k] - i \cdot e_{\delta a_x}[k]$$

$$a_x[k]\, f_{dx}[k] - a_x[k]\, \bar{\bar{f}}_{dx}[k] \approx e_{a_x}[k] \underbrace{\Big\{f_{dx}[k] + (1-\lambda_c) \textstyle\sum_{i=1}^{d} f_{dx}[k-i]\Big\}}_{F_{dx}[k]} - e_{\delta a_x}[k] \underbrace{\Big\{(1-\lambda_c) \textstyle\sum_{i=1}^{d} i \cdot f_{dx}[k-i]\Big\}}_{dF_{dx}[k]} + (1-\lambda$$

$$\boxed{\delta x[k+1] = \lambda_c \cdot \delta x[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \underbrace{\Big\{\delta x_T^d[k] + (1-\lambda_c) n_x[k]\Big\}}_{\varepsilon[k]}}$$

State equation

$$\delta x_1[k+1] = \delta x_2[k]$$
$$\delta x_2[k+1] = \delta x_3[k]$$
$$\delta x_3[k+1] = \lambda_c \cdot \delta x_3[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] - \varepsilon[k]$$
$$a_x[k+1] = a_x[k] + \delta a_x[k]$$
$$\delta a_x[k+1] = \delta a_x[k]$$

Output equation

$$\begin{cases} y_1[k] = \delta x_m[k] = \delta x_1[k] + \underbrace{n_x[k]}_{r_1[k]} \\ y_2[k] = \underbrace{a_x[k-d] + n_a[k]}_{a_{xm}[k]} \approx a_x[k] - d \cdot \delta a_x[k] + \underbrace{n_a[k]}_{r_2[k]} \end{cases}$$

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -d \end{bmatrix}$$
$$\mathbf{y}[k] = \mathbf{H}\mathbf{x}[k] + \mathbf{r}[k]$$

Closed-loop Estimator

$$e_{\delta x_1}[k] = \delta x_1[k] - \delta \hat{x}_1[k]$$

$$e_{a_x}[k] = a_x[k] - \hat{a}_x[k]$$

$$e_{\delta a_x}[k] = \delta a_x[k] - \delta \hat{a}_x[k]$$

$$e_{y_1}[k] = y_1[k] - \hat{y}_1[k] = \delta x_m[k] - \delta \hat{x}_1[k] = e_{\delta x_1}[k] + \underbrace{n_x[k]}_{r_1[k]}$$

$$e_{y_2}[k] = y_2[k] - \hat{y}_2[k] = a_{xm}[k] - \hat{a}_x[k] + d \cdot \delta \hat{a}_x[k] = e_{a_x}[k] - d \cdot e_{\delta a_x}[k] + \underbrace{n_a[k]}_{r_2[k]}$$

$$\delta \hat{x}_1[k+1] = \delta \hat{x}_2[k] + \ell_{11}[k]\, e_{y_1}[k] + \ell_{12}[k]\, e_{y_2}[k]$$
$$\delta \hat{x}_2[k+1] = \delta \hat{x}_3[k] + \ell_{21}[k]\, e_{y_1}[k] + \ell_{22}[k]\, e_{y_2}[k]$$
$$\delta \hat{x}_3[k+1] = \lambda_c \cdot \delta \hat{x}_3[k] + \ell_{31}[k]\, e_{y_1}[k] + \ell_{32}[k]\, e_{y_2}[k]$$
$$\hat{a}_x[k+1] = \hat{a}_x[k] + \delta \hat{a}_x[k] + \ell_{41}[k]\, e_{y_1}[k] + \ell_{42}[k]\, e_{y_2}[k]$$
$$\delta \hat{a}_x[k+1] = \delta \hat{a}_x[k] + \ell_{51}[k]\, e_{y_1}[k] + \ell_{52}[k]\, e_{y_2}[k]$$

Error dynamics

$$
\begin{aligned}
e_{\delta x_1}[k+1] &= e_{\delta x_2}[k] - \ell_{11}[k]\, e_{\delta x_1}[k] - \ell_{12}[k]\, e_{a_x}[k] + \ell_{12}[k] \cdot d \cdot e_{\delta a_x}[k] \\
e_{\delta x_2}[k+1] &= e_{\delta x_3}[k] - \ell_{21}[k]\, e_{\delta x_1}[k] - \ell_{22}[k]\, e_{a_x}[k] + \ell_{22}[k] \cdot d \cdot e_{\delta a_x}[k] \\
e_{\delta x_3}[k+1] &= \lambda_c \cdot e_{\delta x_3}[k] - F_{dx}[k] \cdot e_{a_x}[k] + dF_{dx}[k] \cdot e_{\delta a_x}[k] \\
&\quad - \ell_{31}[k]\, e_{\delta x_1}[k] - \ell_{32}[k]\, e_{a_x}[k] + \ell_{32}[k] \cdot d \cdot e_{\delta a_x}[k] - \underbrace{\varepsilon[k]}_{q_3[k]} \\
e_{a_x}[k+1] &= e_{a_x}[k] + e_{\delta a_x}[k] - \ell_{41}[k]\, e_{\delta x_1}[k] - \ell_{42}[k]\, e_{a_x}[k] + \ell_{42}[k] \cdot d \cdot e_{\delta a_x}[k] \\
e_{\delta a_x}[k+1] &= e_{\delta a_x}[k] - \ell_{51}[k]\, e_{\delta x_1}[k] - \ell_{52}[k]\, e_{a_x}[k] + \ell_{52}[k] \cdot d \cdot e_{\delta a_x}[k]
\end{aligned}
$$

$$
\mathbf{F}_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dx}[k] & dF_{dx}[k] \\
0 & 0 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \begin{bmatrix} 0 & 0 & -\varepsilon[k] & 0 & 0 \end{bmatrix}^\top, \qquad \mathbf{r}[k] = \begin{bmatrix} n_x[k] & n_a[k] \end{bmatrix}^\top
$$

$$
\mathbf{e}[k+1] = \mathbf{F}_e[k]\, \mathbf{e}[k] - \mathbf{L}[k]\, \mathbf{H}\, \mathbf{e}[k] + \mathbf{q}[k] - \mathbf{L}\, \mathbf{r}[k]
$$